ANSWER KEY & MARKING SCHEME

CLASS 8 – CBSE PHYSICS QUESTION PAPER

Maximum Marks: 30

SECTION A: SHORT ANSWER QUESTIONS (8 Marks)

Q1. Force & Pressure

- **Answer:**
    - *(a)* **Atmospheric Pressure:** The force per unit area exerted on a surface by the weight of the atmospheric column of air above it.

SI Unit: Pascal (Pa) or Newton per square meter (N/m^2).

    - *(b)* Pressure is inversely proportional to area ($P = \frac{F}{A}$). Sharp cutting tools have a extremely small surface contact area (A), so even a modest applied force (F) generates high pressure (P) capable of cutting materials easily.

- **Marking Scheme:**
    - **1 Mark:** Correct definition of atmospheric pressure + SI unit (Pa or N/m^2).
    - **1 Mark:** Explaining inverse relation between area and pressure ($P \propto \frac{1}{A}$).

Q2. Sound Wave Terminology

- **Answer:**
    - *(a)* **Amplitude:** The maximum displacement of a vibrating particle from its mean position. Loudness is directly proportional to the square of amplitude (Loudness $\propto$ Amplitude^2).
    - *(b)* **Frequency:** The number of complete wave oscillations per second. Its SI unit is **Hertz (**Hz**)**. High frequency corresponds to high pitch (shrill sound), whereas low frequency corresponds to low pitch (hoarse/flat sound).

- **Marking Scheme:**
    - **1 Mark:** Amplitude definition + square law relation to loudness.
    - **1 Mark:** Frequency definition + SI unit (Hz) + pitch relationship.

Q3. Friction Types & Mechanisms

- **Answer:**
    - **Increasing Order:** Rolling Friction $<$ Sliding Friction $<$ Static Friction.
    - **Interlocking of Irregularities:** The physical meshing together of microscopic peaks and valleys present on contacting rough surfaces, resisting relative motion.
    - **Lubricants:** Substances (like oil/grease) that form a thin layer between moving surfaces, smoothing out irregularities and replacing solid-to-solid friction with fluid friction.

- **Marking Scheme:**
    - **1 Mark:** Correct ordering of friction magnitudes.

- **1 Mark:** Defining interlocking + explaining lubricant action.

Q4. Chemical Effects of Electric Current

- **Answer:**
  - **Electroplating:** The deposition of a superior metal layer over another conductive object via electrolysis.
  - **Electrodes:** Metallic rods/plates immersed in an electrolyte through which electric current enters or leaves the solution.
  - **Setup:**
    - *Electrolyte:* Copper Sulphate solution ($CuSO_4$).
    - *Cathode (Negative Electrode):* The iron object to be plated.
- **Marking Scheme:**
  - **1 Mark:** Accurate definitions of electroplating and electrodes.
  - **1 Mark:** Correct electrolyte ($CuSO_4$) and cathode (iron object) identification.

SECTION B: MEDIUM ANSWER QUESTIONS (12 Marks)

Q5. Optics & Reflection Analysis

- **Answer:**
  - *(a)* **Laws of Reflection:**
    1. The angle of incidence (θ_i) is always equal to the angle of reflection (θ_r).
    2. The incident ray, the reflected ray, and the normal at the point of incidence all lie in the same plane.
  - *(b)* Given glance angle with surface $g = 35°$:

Since Normal $\perp$ Surface $= 90°$:

$$\theta_i = 90° - 35° = \mathbf{55}°$$

According to the Law of Reflection ($\theta_i = \theta_r$):

$$\theta_r = \mathbf{55}°$$

- **Marking Scheme:**
  - **1.5 Marks:** Stating both laws of reflection clearly.
  - **1.5 Marks:** Correct normal calculation obtaining $\theta_i = 55°$ and $\theta_r = 55°$.

Q6. Electrostatics & Instrumentation

- **Answer:**
  - *(a)* **Conductors:** Materials with high free-electron mobility that readily allow charge flow (e.g., Copper, Iron).

Insulators: Materials lacking free electrons that restrict electric charge flow (e.g., Rubber, Glass).

  - *(b)* **Gold-Leaf Electroscope Working:** When a charged object contacts the brass cap, charges conduct down the metal rod to two thin gold leaves. Acquiring like charges, the leaves experience **electrostatic repulsion** and diverge.

- **Marking Scheme:**
    - **1.5 Marks:** Conductors vs. insulators distinction based on free electrons + examples.
    - **1.5 Marks:** Explaining electroscope charge transfer and leaf repulsion.

Q7. Acoustics & Noise Pollution

- **Answer:**
    - *(a)* **Audible Spectrum:** Human ear perceives acoustic frequencies between 20 Hz **and** $20{,}000$ Hz **(**20 kHz**)**.
        - *Infrasonic:* Frequencies below 20 Hz.
        - *Ultrasonic:* Frequencies above $20{,}000$ Hz.
    - *(b)* Prolonged exposure to high-decibel noise disrupts cardiovascular balance leading to elevated blood pressure (**hypertension**) and auditory nerve distress resulting in continuous ringing in ears (**tinnitus**).
- **Marking Scheme:**
    - **2 Marks:** Exact frequency bounds (20 Hz $-$ 20 kHz) + definitions of infra/ultrasonic.
    - **1 Mark:** Explaining health impacts (hypertension & tinnitus).

Q8. Fluid Mechanics & Hydrostatic Pressure

- **Answer:**
    - *(a)* **Hydrostatic Pressure:** The pressure exerted by a fluid at rest due to the force of gravity.

Factors ($P = h \cdot \rho \cdot g$**):**

1. Depth of the liquid column (h).
2. Density of the liquid (ρ).

- *(b)* Fluid pressure increases directly with ocean depth ($P = h\rho g$). Reinforced pressure suits prevent the overwhelming external hydrostatic pressure from collapsing a diver's lungs and body cavities.

- **Marking Scheme:**
    - **2 Marks:** Hydrostatic pressure definition + listing two governing factors (h, ρ).
    - **1 Mark:** Applying $P \propto h$ to justify reinforced suits for deep-sea divers.

SECTION C: LONG ANSWER & ANALYTICAL PROBLEMS (10 Marks)

Q9. Wave Motion & Numerical Computations

- **Answer:**
    - *(a)* **Sonar Seabed Depth Computation:**

Given: $v = 1500$ m/s, $t = 2.4$ s

Formula: $d = \frac{v \times t}{2}$

$$d = \frac{1500 \times 2.4}{2} = 1500 \times 1.2 = \mathbf{1800} \text{ meters}$$

- *(b)* **Oscillation Parameters:**

Given: $n = 1200$ oscillations, $t = 15$ seconds

1. **Frequency (f):**

$$f = \frac{\text{Oscillations}}{\text{Time}} = \frac{1200}{15} = \mathbf{80}\ \text{Hz}$$

2. **Time Period (T):**

$$T = \frac{1}{f} = \frac{1}{80} = \mathbf{0.0125}\ \text{seconds}$$

- **Marking Scheme:**
  - **2 Marks:** Correct echo depth computation (1800 m).
  - **1.5 Marks:** Correct frequency computation (80 Hz).
  - **1.5 Marks:** Correct time period computation (0.0125 s).

Q10. Human Eye Anatomy & Refractive Defects

- **Answer:**
  - *(a)* **Eye Components:**
    1. **Cornea & Lens:** Cornea provides major fixed light refraction; the crystalline lens adjusts focal length via ciliary muscles to focus sharp images on the retina.
    2. **Iris & Pupil:** Iris acts as an adjustable diaphragm controlling pupil aperture size to regulate light flux entering the eye.
    3. **Retina & Optic Nerve:** Retina houses photoreceptor cells (**rods** for light intensity, **cones** for color perception); optic nerve transmits generated electrical impulses to the brain.
  - *(b)* **Vision Defects:**
    - **Myopia (Nearsightedness):** Distant light rays converge **in front of the retina** (due to elongated eyeball or high lens curvature). Corrected using a **diverging (concave) lens**.
    - **Hypermetropia (Farsightedness):** Near light rays converge **behind the retina** (due to shortened eyeball or low lens curvature). Corrected using a **converging (convex) lens**.
- **Marking Scheme:**
  - **3 Marks:** Clear functional breakdown of all three eye component pairs (1 mark each).
  - **2 Marks:** Image position placement for myopia (in front) vs hypermetropia (behind) + correct corrective lenses specified (concave vs convex).